

Asymmetric Encryption (Public Key Cryptography)

Teacher: Dr. Uzma Jamil

1. Introduction

What is Encryption?

- Encryption = converting **readable data (plaintext)** into **unreadable form (ciphertext)**
- Purpose:
 - Protect confidentiality
 - Secure communication

Quick Recall

- **Symmetric Encryption** → one shared key
- Problem: How to safely share the key?

This leads to **Asymmetric Encryption**

2. Core Concept of Asymmetric Encryption

Definition

Asymmetric Encryption uses **two different but related keys**:

- **Public Key** (shared with everyone)
- **Private Key** (kept secret)

How it Works (Simple Idea)

- If something is encrypted with **public key** → **only private key can decrypt**
- If something is encrypted with **private key** → **public key can verify**

3. Working Mechanism (10 minutes)

Step-by-Step Process

Case 1: Confidential Communication

1. Sender gets receiver's **public key**

2. Sender encrypts message
3. Receiver decrypts using **private key**

Case 2: Authentication (Digital Signature)

1. Sender encrypts message (or hash) with **private key**
2. Receiver uses **public key** to verify

4. Visual Understanding (Conceptual Flow)

Encryption:

Message → Public Key → Ciphertext → Private Key → Original Message

Signature:

Message → Private Key → Signature → Public Key → Verified

5. Real-Life Example (10 minutes)

Example: Locked Mailbox System

Imagine:

- You have a **mailbox with a slot (public key)**
- Anyone can drop a letter inside
- But only **you have the key (private key)** to open it

This ensures:

- Anyone can send messages
- Only you can read them

Practical Example: HTTPS (Secure Websites)

When you visit a website:

- Your browser gets website's **public key**
- Data is encrypted
- Website decrypts using **private key**

That's why you see  in browser

6. Common Algorithms

1. RSA (Most Popular)

- Based on **large prime numbers**
- Widely used in:
 - Secure email
 - SSL/TLS

2. ECC (Elliptic Curve Cryptography)

- More efficient than RSA
- Used in:
 - Mobile devices
 - Modern security systems

3. Diffie-Hellman (Key Exchange)

- Used to **securely share keys**
- Not directly for encryption

7. Advantages and Limitations

Advantages

- No need to share secret key
- Enables **secure communication**
- Supports **digital signatures**

Limitations

- Slower than symmetric encryption
- Requires more computation
- Complex implementation

8. Hybrid Encryption (Important Concept)

Real systems use **both types**

How?

1. Asymmetric encryption used to share key
2. Symmetric encryption used for data

Example:

- TLS/HTTPS uses:
 - RSA/ECC → key exchange
 - AES → data encryption

9. Applications

- Secure websites (HTTPS)
- Email encryption (PGP)
- Digital signatures
- Cryptocurrency (Blockchain)
- Secure login systems

10. Small Activity / Class Engagement

Activity:

Ask students:

- If you want to send a secret message to a teacher:
 - Which key will you use?

Answer: Teacher's **public key**

Quick Quiz

1. How many keys are used in asymmetric encryption?
Two

2. Which key is shared publicly?
Public key
3. Which key must remain secret?
Private key

11. Summary

- Asymmetric encryption uses **two keys**
- Solves **key distribution problem**
- Used in:
 - Secure communication
 - Authentication
- Works with symmetric encryption in real systems

12. References

- William Stallings – *Computer Security: Principles and Practice*
- NIST (National Institute of Standards and Technology)
- OWASP (Open Web Application Security Project)